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High Performance Ecoinformatics

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MSc Computer Science
Data Science and Engineering Curriculum

High Performance Econinformatics

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DEDICATION

Abstract Ecoinformatics is a modern method of doing science that enables scientists to generate new knowledge through innovative tools and approaches. Ecology is increasingly becoming a data-intensive science, involving the acquisition, preservation, and analysis of vast amounts of data from a variety of sources, including satellite and airborne remote sensing, instruments, sensors, and human observations. Ecologists are increasingly addressing larger-scale questions of both scientific and societal relevance, such as climate change and its consequences for ecosystems and related ecosystem services in the future. The topic of this thesis project stems from a research study conducted by the Department of Earth, Environmental, and Life Sciences (DISTAV, University of Genoa) that analyzes the shift in the distribution of natural habitats in the Alps due to future climate change to support decision-making processes regarding conservation planning and prioritization. The analysis is conducted using Species Distribution Models (SDMs), a dataset of approximately 3 million data points, and future climate change scenarios based on General Circulation Models (climate models). The statistical software R (<https://www.r-project.org/>) is used to generate future predictions of habitat distribution changes and biodiversity loss. The thesis candidate will work on improving the script and algorithmic functions, written in R, to streamline model analysis based on their knowledge and computer science background. This thesis project leverages interdisciplinary collaboration between DIBRIS and DISTAV to achieve better scientific results. Depending on the candidate's interest and quality of results, the thesis can be upgraded with asterisk.

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Chapter 1

Introduction and Motivation

introduzione deve essere scritta alla fine

1.1 Project context (taken from **Methods_GC_LD.pdf**)

Thesis proposal, ecoinformatics, Lucia.

Eco-informatics is a modern way of conducting scientific research that enables scientists to generate new knowledge through innovative tools and approaches. Ecology is increasingly becoming a data-intensive science that involves acquiring, storing, and analysing large amounts of data from a range of sources, including satellite and aerial remote sensing, instruments, sensors, and human observations. Ecologists increasingly address large-scale questions with both scientific and social relevance, such as climate change and its future effects on ecosystems and the ecosystem services associated with them. The topic of this thesis project originated from a research study conducted by the Department of Earth, Environmental and Life Sciences (DISTAV, University of Genoa). The study analyses changes in the distribution of natural habitats in the Alps caused by future climate change in order to support decision-making on conservation planning and prioritisation. The analysis uses Species Distribution Models, a dataset of approximately four million records, and future climate-change scenarios based on General Circulation Models. The statistical software R (<https://www.r-project.org/>) is used to generate predictions of future changes in habitat distribution and biodiversity loss. The thesis student will improve the working script and its algorithmic functions to make the model analysis more efficient, drawing on their knowledge and studies in computer science. This thesis project involves interdisciplinary collaboration between DIBRIS and DISTAV, which can help produce better scientific results.

Chapter 2

Background

real world and eventually related work

2.1 Ecoinformatics

2.2 Biological study context

An internal DISTAV methods document describes the biological study supported by the computational workflow. The study examines how climate change may alter the suitability of Alpine grassland habitats. It considers ten dry, alpine and subalpine habitat types from the EUNIS classification and 167 diagnostic plant species. Occurrence records are associated with environmental predictors at a spatial resolution of approximately 1 km. The predictors describe climate, soil and topographic heterogeneity. Species Distribution Models combine five modelling techniques and ensemble forecasts to estimate present and future habitat suitability.

The biological protocol also defines the analyses that follow model projection: changes in suitability, spatial shifts, habitat-level hotspot and cold-spot maps, and the relationship between suitability changes and species tolerance. This document provides the scientific context for the computing work, but it is not treated as the exact specification of every run. Dataset sizes, model repetitions and available climate scenarios changed during the project; the computational chapters therefore report the configuration recorded by the corresponding scripts and run manifests.

2.3 Data used

2.4 Related work

Chapter 3

Code as is

Existing workflow and computational problem.

3.1 R

3.2 BIOMOD2

3.3 Code pipeline

3.4 Original execution times

3.5 Improved execution times

3.6 Preliminary measurements on Spartaco

Before the Leonardo experiments, Lucia manually recorded four trial runs on Spartaco. The worksheet records per-phase time and RAM observations for two species, with different pseudo-absence settings and CPU counts. Table 3.1 preserves selected values from those records.

The measurements are useful for reconstructing the starting point of the project. They are not a controlled benchmark: the worksheet does not identify the code revision, container,

hardware configuration, input version or measurement method. The source label `n_cells` is retained in the table because its meaning cannot be established from the worksheet alone. These results are therefore not compared directly with the later Leonardo measurements.

Table 3.1: Selected fields from four manually recorded trial runs on Spartaco. `n_cells` retains the source label. Time units are those recorded in the original worksheet; the maximum RAM is the largest phase value in that row.

Species	<code>n_cells</code>	PA	Rep.	RUN	CPU	Formatting	Models	Future projection	Max RAM (GB)
<i>Omalotheca hoppeana</i>	681	10000	10	10	5	1.9 h	11.33 min	3.65 min	52.9
<i>Omalotheca hoppeana</i>	681	10000	10	10	10	2.02 h	7.74 min	8.07 min	56.16
<i>Agrostis capillaris</i>	100000	5000	5	5	5	32.25 min	53.87 min	3.73 min	26
<i>Agrostis capillaris</i>	100000	10000	5	5	5	59.37 min	1.6 h	4.22 min	29.96

Chapter 4

My new code

Design and implementation of the improvements.

4.1 Container

4.2 Parallel model execution

4.3 Parallel I/O

Chapter 5

Experimental results

Experimental methodology and results.

5.1 Leonardo

5.2 Tests

5.3 Discussion and lessons learned

Chapter 6

Conclusions and future development

6.1 Conclusions

6.2 Future development

6.3 Ecoinformatics

What it is.

Why large datasets are used.

Why high-performance computers are used.

Purpose.

6.4 Data used

In the **Eval...** matrix, results with **calibration** below 0.5 are considered less performant.

6.5 Computer science objective

To be defined.

6.6 Interdisciplinary collaboration

Optional section.

Chapter 7

Appendix

Temporary or optional chapter.

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